Direct device access from the SmartNIC towards datacenter disaggregation

State-of-the-art

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Abstract

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This paper an is draft aiming at analyzing state-of-the-art solution for DDA from the smartNIC towards datacenter disaggregation and is not made to be submitted anywhere

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1 Introduction

For decades, servers in datacenters were designed under a monolithic model, meaning that all hardware resources are physically attached to the same frame. This model is now a bottleneck for performance, network throughput and to efficiently consume resources. In order to solve the monolithic challenge, a new architectural design has been proposed : disaggregation.

2 Definitions

2.1 Disaggregation

Disaggregation is a new model where hardware resources such as computing power, memory, accelerators, storage, ... are divided into pools (or nodes) interconnected into each other via a high-speed network. This allows resources to be allocated on demand, with better utilization and availability. There is mainly two levels of disaggregation.
Partial disaggregation separates storage components from CPU and memory which remain integrated together. This level of disaggregation has already been widely implemented since early 2020s.
Fully disaggregation regroups same type of resource into clusters which are interconnected over a network to allow communication and data exchange between them. [1]

2.2 RDMA

Remote Direct Memory Access (RDMA) is a protocol that allows to access memory from one computer into another computer without involving operating system [2]. This protocol offers high throughput and low latency, and will be referred from many articles covering this topic.

2.3 Zero-copy

Zero-copy enable the possibility to transfer data from one device to another without requiring CPU to copy the data. This avoids redundant copies, reducing CPU usage.

2.4 SPDK

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2.5 NVMe-oF

NVMe over-Fabric (NVMe-oF) is an extension of NVMe command set. It allows to send these commands over networked fabrics [3], [4], [5]

3 Architectural challenges

Challenges whose origin is architectural design can be divided into two classes :

- Problems due to traditional monolithic architectural server. This type of problem is the core reason of why datacenters are moving from monolithic architecture from disaggregated one.

- New challenges brought by disaggregation. Disaggregated computing is a relatively new design in system and datacenter who breaks many of assumption previously made in the research. Researchers and datacenter operators need to find creative solutions to outcome these challenges and enjoy the benefits of disaggregation.

3.1 Monolithic challenges

The main drawback with monolithic architecture (MA) is **resource stranding**. When a server exhausts one of its resource, for example memory, all other resources cannot be utilized and result in a waste of resources.
Another non-negligible issue concerning MA is failure tolerance and availability. When a piece of hardware fails this may cause the entire server to fails and become unavailable.
Finally, replacing or upgrading hardware in MA occurs maintenance that can be costly.

3.2 Disaggregation challenges

One of the main challenge in disaggregated architecture (DA) is that network bandwidth becomes a central point of communication between clusters; CPU-storage communication needs to be done in a few milliseconds while CPU-memory needs to be done in a few nanoseconds, usually 100ns [1].
Existing network stacks face significant trade-offs in terms of flexibility and performance. For example, RDMA and RDMA over Convergent Ethernet (RoCE) are high-performance protocols, but they lack flexibility because they are tightly coupled to specific hardware, making them difficult to adapt. In addition to limited flexibility, RDMA suffers from issues like head-of-line blocking, congestion, and vendor lock-in, due to hardware requirements that aren't met by commodity equipment. On the other hand, network stacks like Linux TCP offer greater flexibility but at the cost of poor performance and inefficient CPU utilization [6].

4 State-of-the-art designs and solutions

Some hardware pieces are evolving, like storage moving from hard disk drive to NVMe SSD and RAM becoming less expensive. This allows researchers and enterprises to discover several new ways to achieve high performance computing. In this section we'll review main contributions from the literature.

4.1 Zero-copy data path

Skiadopoulos et al. [6] designed a new prototype called *ZeroNIC* which can zero-copy data between NIC and accelerators. Its main advantage is that is it agnostic of the accelerator type and canrun on CPU, GPU, FPGA, ... NIC hardware split header and payload, transferring the latter directly to the receiver applications buffers without intermediates copies.
Concerning NVMe storage, Sun et al. [7] focused on NVMe-oF target offloading which extends zero-copy mechanism, transferring data to the DPU's host channel adapter (HCA) via peer-to-peer PCI communication (P2PDMA) and thus creating a new kind of zero-copy data path.

4.2 Batched write

Batched write mechanism retains write operations until an arbitrary threshold is achieved, then all these operations are run consecutively while a group commit ensures atomicity of operations. The goal is to improve throughput and reduce the accelerator CPU load. While it is not technically a contribution brought by disaggregation, its evolution into such systems is still relevant to be mentioned.

4.3 Split stacks

Split stacks are networking stacks divided in several planes, each one optimized for a specific task. IO-TCP from Kim et al. [8] separate the traditional Linux TCP stack into a stack running on the host CPU, performing control operations, and thus called the control plane. Such operations are complex and benefits from features of the CISC CPU (for example branch prediction, vectorized instructions, ...). The host stack extends the mTCP

[9] API with methods to offload file opening, writing and closing, such that applications have the most minimal changes to do in order to run itself on IO-TCP.
The data plane handles all operations related to data packet preparation and transfer. These operations are simpler in comparison of the control plane and then fit better the ARM cores of the SmartNIC. They also benefit from being stateless which suits the usage of the smartNIC. File and disk I/O are offloaded to otherwise it would interfere with the control plane on host CPU. Data plane operations are virtually performed and enable a mechanism of SEND / ECHO which translate in multiple MTU packets transfer.
Many difficulties due to the split introduce themselves, like RTT measurement, retransmission and error handling, but are essentially resolved with the set of commands provided by the implementation.

5 Unresolved challenges

There are some challenges still left open upon reading papers in **section 4**. This section aims to list them, with the goal to define a clear approach for working on the master's thesis as well as a precise problem to solve. IO-TCP [8] lacks of dynamic synchronization. This because authors assumes that files don't change during the content delivery service. In their solution, both host and smartNIC share the same file system, but any change on the host-side FS isn't propagated to the SmartNIC stack.
Another problem is that file reading on NVMe disks is done with direct I/O on a regular ext4 file system. SPDK reaches the best performance, but support for file system isn't very developed for now.
Next, with a smaller amount of connection (typically < 4), Linux TCP stack outperforms IO-TCP by over 1.5 times.

Batched writes used in Scalio [7] to improve throughput of write operations come at the cost of latency increase. This can lead to non-negligible issues in environment where low latency is critical.

6 Objectives

This paper will focus on fully disaggregated datacenter, especially on interaction between NVMe storage pools and SmartNIC. One of our objectives is to allow SmartNICs to communicate directly with the storage without involving any compute pool.

7 Conclusion

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